

Ch. 8 — P01 (T) (s,S) Inventory Manual Trace

Problem Bank | Chapter 8 | Type: T | Difficulty:

Chapter: 8 Type: T Difficulty:

A warehouse uses an $(s, S) = (5, 20)$ continuous-review inventory policy. The initial inventory level is 20 units. Orders are received instantaneously (zero lead time). Use the demand events below:

Event #	Time	Demand (units)
1	2	4
2	5	3
3	9	5
4	11	2
5	14	6
6	18	4
7	21	3
8	24	5

Tasks

- Complete the trace table. For each event, record:
 - Time
 - Demand
 - Inventory *before* demand
 - Inventory *after* demand
 - Was an order placed? (Yes/No)
 - Inventory *after* receiving order (if applicable)
- Draw the inventory-over-time sawtooth plot (hand-sketch acceptable).
- Compute the **time-average inventory level** over $[0, 24]$.
- Identify all time intervals when a **stockout** (inventory ≤ 0) would have occurred if the policy had been $(s, S) = (3, 15)$ instead.

5. What is the total number of orders placed under the $(5, 20)$ policy during $[0, 24]$?